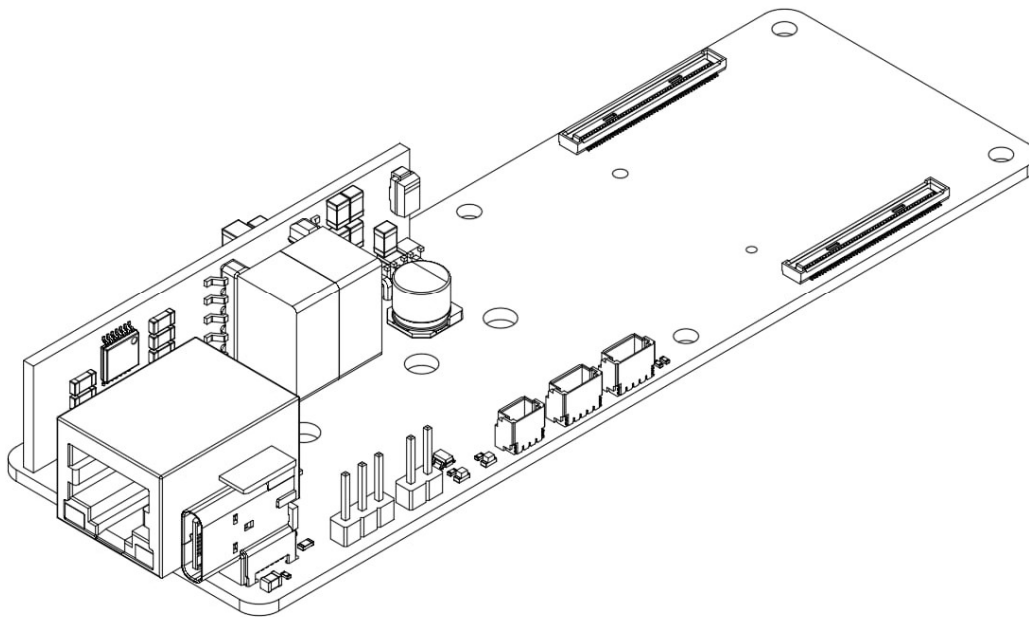


CM4/CM5

Carrier Board Rev1

Technical Datasheet

Compact network-attached compute carrier for Raspberry Pi CM4-, CM5- and compatible compute modules.



120 mm × 40 mm • Gigabit Ethernet • 25 W PoE • USB-C • M.2 M-Key

Overview

The CM4/CM5 Carrier Board Rev1 is a compact, headless embedded-computing platform intended for edge computing, home-lab services, network appliances and industrial prototypes. Two 100-pin Hirose connectors accept Raspberry Pi Compute Module 4 and Compute Module 5 modules and compatible third-party modules.

Key specifications

- Six-layer PCB in a 120 mm × 40 mm form factor.
- USB-C for USB 2.0 data and 5 V power input.
- Gigabit Ethernet RJ45 MagJack with integrated link/activity indicators.
- 25 W PoE powered-device input through a companion PoE module.
- Jumper-selectable USB-C or PoE power source.
- M.2 M-Key socket supporting 2230, 2242 and 2260 modules.
- microSD socket for Compute Module Lite variants.
- UART, I²C/Qwiic and four-pin PWM fan connectors.
- PWR and ACT indicator LEDs and USB boot/recovery header.

Power architecture

USB-C provides the board with 5 V at up to 3 A using the Type-C CC pull-down configuration. The PoE path accepts 44–57 V through Ethernet and provides an isolated regulated 5 V output through the companion 25 W PoE module. A three-pin selector header connects the system rail to either USB-C or PoE and prevents the two inputs from being tied together.

The carrier distributes the 5 V input to the Compute Module and peripherals and provides a dedicated 3.3 V rail for M.2 devices. The red PWR LED indicates the presence of the 5 V rail. The green ACT LED follows Compute Module storage/system activity.

Power selection: Configure the selector jumper only while the board is powered off.

Connectors and interfaces

USB-C

The USB Type-C receptacle carries VBUS, GND, D+ and D– for combined 5 V input and USB 2.0 device connectivity. USB gadget mode can expose the Compute Module as a USB Ethernet interface when connected to a host.

Gigabit Ethernet and PoE

The right-angle RJ45 MagJack supports 10/100/1000BASE-T Ethernet and includes link/activity LEDs. PoE power and control signals are routed to an eight-pin footprint for the companion PoE daughterboard, with additional ESD protection at the interface.

UART debug

The three-pin JST header exposes 3.3 V TTL TXD, RXD and GND for boot logs, Linux console access and board bring-up through a USB-to-UART adapter.

I²C / Qwiic

The keyed four-pin JST-SH connector carries GND, 3.3 V, SDA and SCL. It follows the Qwiic pin arrangement and supports 3.3 V I²C peripherals and daisy-chained devices.

PWM fan

The four-pin JST-SH connector follows the Raspberry Pi 5 fan pinout: 5 V, GND, PWM control and tachometer feedback. CM5 provides PWM fan control. With CM4-compatible modules in Rev1, PWM control is unavailable and the fan operates at full speed while powered.

Third-party modules: Some modules marketed as CM5 use a CM4-compatible architecture for power delivery or fan control. Verify the module documentation before use.

Storage and configuration

microSD

The push-push microSD socket is intended for CM4 Lite, CM5 Lite and compatible modules without onboard eMMC. Standard Compute Modules with eMMC do not use this socket.

M.2 M-Key

The M.2 connector accepts 2230, 2242 and 2260 M-Key cards. Typical devices include NVMe SSDs, PCIe AI accelerators and interface adapters. PCIe Gen 2 ×1 operation is the supported default. PCIe Gen 3 ×1 may be possible but is experimental and unsupported.

Rev1 limitation: Raspberry Pi CM4 and Radxa CM5 modules cannot access the M.2 interface on this board revision. This limitation was corrected in Rev2.

Jumper settings

Pins	Function	Description
1–2	USB power	Connect the system rail to the USB-C 5 V input.
2–3	PoE power	Connect the system rail to the PoE daughterboard.
BOOT	USB recovery	Fit during power-up to enter USB boot/recovery mode.

USB boot / recovery

Fit the BOOT jumper before applying power to force USB boot mode for eMMC flashing, recovery or manufacturing. Remove the jumper for normal boot from the configured storage device.

Using an NVMe SSD

The following procedure applies to supported CM5-based configurations. Install a compatible 2230, 2242 or 2260 NVMe drive and confirm that the operating system includes the appropriate driver.

1. Enable PCIe

Open the boot configuration file:

```
sudo nano /boot/firmware/config.txt
```

Add the PCIe parameter, save the file and reboot:

```
dtparam=pciex1
sudo reboot
```

If required for compatibility, force PCIe Gen 2:

```
dtparam=pciex1_gen=2
```

2. Confirm device detection

Use the following commands after reboot:

```
lsblk
lspci
lsblk -o NAME,SIZE,MODEL,TYPE,MOUNTPOINT
```

A detected drive normally appears as nvme0n1 in lsblk and as a Non-Volatile memory controller in lspci. Existing partitions appear as nvme0n1p1, nvme0n1p2 and similar entries.

3. Check for power throttling

```
vcgencmd get_throttled
```

A result of throttled=0x0 indicates that no throttling flags are currently set. Any other result may indicate a present or historical power-related condition.

Data safety: Back up important data before partitioning or formatting an SSD.

Mechanical layout

The carrier PCB measures 120 mm × 40 mm. The diagram below provides top, bottom and side views with principal component and mounting dimensions in millimetres.

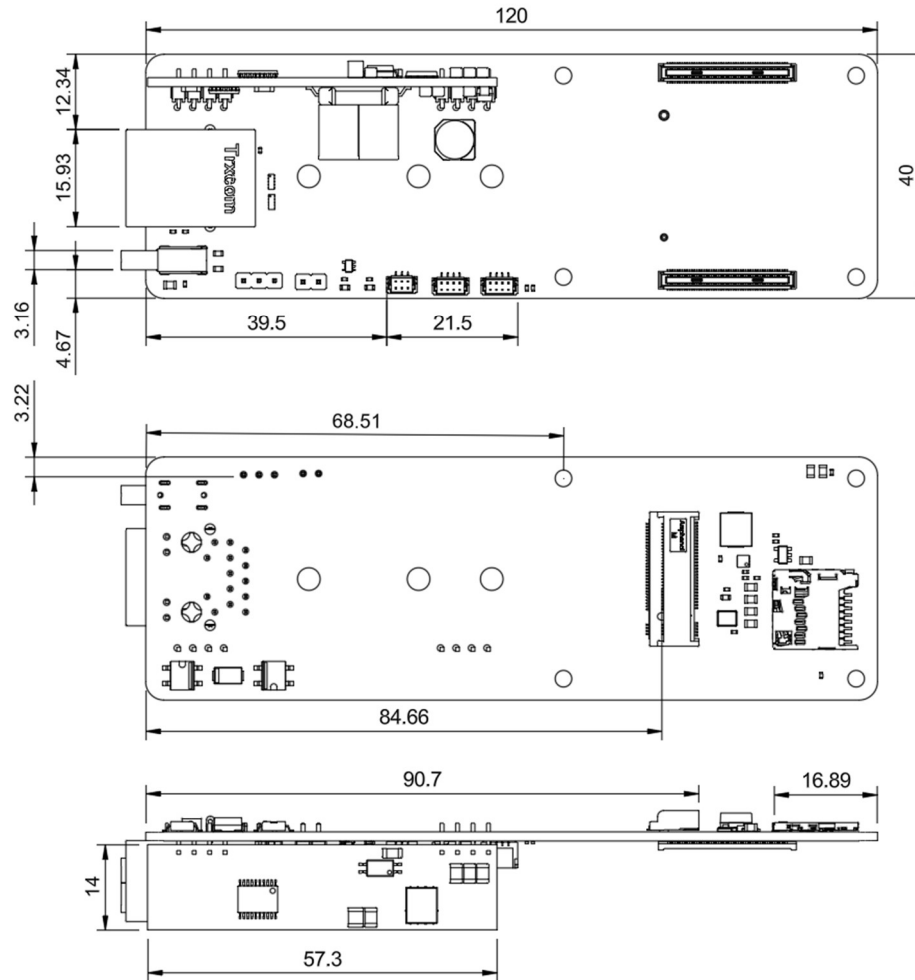


Figure 1. Mechanical outline and principal dimensions.

Document information

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Board revision	Rev1
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